

Eyes in Detail

Glaucoma (intraocular pressure and optic nerve), macular degeneration, cataracts, myopia and dry eye

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Unfavourable. The two strong signs are glaucoma and myopia. In the primary open angle glaucoma, the scores are in the 97.9, 81.5 and 59.2 (genomic) percentiles, and the loci load at percentile 91.8. The main cause is the intraocular pressure: score at 94.0 and loci load at 93.4. The vulnerability of the optic nerve (CDKN2B-AS1 A/A, SIX6 H141N C/C and TXNRD2 C/C, 2 copies each) and **GAS7 A/A** are joined. In the myopia, the genomic diagnosis is at percentile 99.2, the loci load at percentile 99.5 and the refractive error in the myopic side (percentile 7.6), with KCNQ5 and PRSS56 A/A. The cataracts are in 86.6 (genomic) and 72.2.

Favourable. The macula is protected: **ARMS2 A69S G/G** (without the highest risk allele), CETP without the risk allele, and typical DMI scores (53.4, 71.7 and 41.1) with loci load at 32.7. **MYOC Q368X** is absent, **LOXL1** has 1 protective allele (G/A) and TMCO1 and AFAP1 have no risk alleles. The excavation of the optic nerve is typical (loci at 38.1).

What does it mean? Glaucoma is the ocular priority. With the polygenic risk in the upper decile, genetically high intraocular pressure and myopia (which, by itself, increases the risk of glaucoma), early detection makes all the difference: glaucoma does not give symptoms until the loss of visual field, which is irreversible, and early treatment stops it. This confirms \$46. CFH Y402H C/T (1 copy) is compensated by ARMS2 G/G.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality DR2 ≥ 0.8 . Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 25 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 22 of the chip, 3 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 12 PGS Catalog scores calculated locally with 4,473,359 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2384 loci, 2160 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In the scores of disease, a higher percentile = more risk. In intraocular pressure, a higher percentile = higher pressure. In the refractive error, a lower percentile = more myopic. The scores marked as already used keep the values of §46.

Glaucoma

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Primary open angle glaucoma	PGS002741 (Waksmunski 2022, 127 variants)	121 127 (weighted coverage 90%)	97.9	MORE LIKELY	unfavourable
Glaucoma	PGS000137 (Craig 2020, 2 673 variants)	2,385 2,673 (weighted coverage 84%)	81.5	MORE LIKELY	unfavourable
Primary open angle glaucoma (genomic)	PGS001797 (Wang 2023, PRS-CS, 885 thousand variants)	860,137 885,417 (weighted coverage 96%)	59.2	TYPICAL	typical
Intraocular pressure	PGS000879 (Qassim 2020, 146 variants)	144 146 (weighted coverage 98%)	63.3	TYPICAL	typical
Intraocular pressure	PGS000124 (MacGregor 2018, 103 variants)	103 103 (weighted coverage 100%)	94.0	HIGH	unfavourable

- **Primary open angle glaucoma:** In §46 (Vision and Hearing).
- **Glaucoma:** Already in §46.

Macula

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Macular degeneration of age	PGS004952 (Herold 2023, 52 variants)	46 52 (weighted coverage 70%)	53.4	TYPICAL	typical
Macular degeneration of age	PGS002269 (Zekavat 2022, 47 variants)	42 47 (weighted coverage 72%)	71.7	TYPICAL	typical
Age macular degeneration (genomic)	PGS004606 (Gorman 2024, PRS-CS, 1 million variants)	984,872 1,000,946 (weighted coverage 98%)	41.1	TYPICAL	typical

- **Macular degeneration of age:** Already in §46.

Crystalline

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Cataracts	PGS001837 (Privé 2022, 11 617 variants)	11,268 11,617 (weighted coverage 97%)	72.2	TYPICAL	typical
Cataracts (genomic)	PGS005149 (Jung 2025, PRS-CSx, 908 thousand variants)	886,908 908,465 (weighted coverage 97%)	86.6	MORE LIKELY	unfavourable

- **Cataracts:** Already in §46.

Myopia and refractive error

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Refractive error (spherical equivalent)	PGS003860 (Clark 2023, LDpred2, 768 thousand variants)	754,706 767,867 (weighted coverage 97%)	7.6	MORE MIOPICO	without judgment
Diagnosis of genomic myopia	PGS003564 (Ding 2023, LDpred2, 980 thousand variants)	972,627 979,739 (weighted coverage 99%)	99.2	MORE LIKELY	unfavourable

- **Refractive error (spherical equivalent):** In §46. Low percentile : myopic trend.

4. Key variants by topic

Variants of drainage of aqueous mood and intraocular pressure, vulnerability of ganglion cells, complement and lipids of the macula, lens and axial growth of the eye. The alleles with effect published in the protective sense were reversed to the sense of risk.

Glaucoma: intraocular pressure and drainage

Trabecular mesh and Schlemm channel genes that regulate the output of aqueous mood.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TMC01	rs4656461		A/A	chip	G (0 copy/s)	favourable	Calcium channel of RE in the trabecular mesh: the strongest common locus of intraocular pressure [GWAS Catalog: Intraocular pressure, G allele, beta 0.291 [0.25-0.33] unit increase, p=1E-43, PMID 29617998]	Strong
CAV1/CAV2	rs4236601		G/A	chip	A (1 copy/s)	unfavourable (light)	CAV1/CAV2 (caveolins of aqueous mood drainage): Allele A rises intraocular pressure and glaucoma (~1.3×); G lowers [GWAS Catalog: Intraocular pressure, allele G, beta -0.22]	Moderate
ABCA1	rs2472493		G/A	chip	G (1 copy/s)	unfavourable (light)	Cholesterol transporter in the trabecular mesh and in the retina [GWAS Catalog: Intraocular pressure, G allele, beta 0.207 [0.18-0.23] unit increase, p=6E-59, PMID 29785010]	Moderate
GAS7	rs9913911		A/A	chip	A (2 copy/s)	unfavourable	GAS7 locus of intraocular pressure [GWAS Catalog: Primary open angle glaucoma (multi-trait analysis), allele A, beta 0.155 [0.14-0.17] unit increase, p=6E-83, PMID 37386247]	Moderate
AFAP1	rs4619890		A/A	chip	G (0 copy/s)	favourable	Locus AFAP1 (cytoskeleton) of glaucoma [GWAS Catalog: Glaucoma (primary open-angle), allele G, OR 1.20, p=7E-10, PMID 25173105]	Moderate
MYOC	rs74315329	Q368X	G/G	chip	A (0 copy/s)	favourable	Q368X: the most common high-effect variant of juvenile/adult glaucoma (penetration ~50-70%)	Strong
LOXL1	rs3825942	G153D	G/A	chip	G (1 copy/s)	favourable	G153D: G allele is associated with pseudoexfoliation syndrome and exfoliative glaucoma	Moderate

Glaucoma: vulnerability of the optic nerve

Genes that decide how much the optic nerve can withstand a given pressure.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CDKN2B-AS1 (9p21)	rs4977756		A/A	chip	A (2 copy/s)	unfavourable	Locus 9p21 (ANRIL): death of retinal ganglion cells; normal pressure glaucoma [GWAS Catalog: Glaucoma (primary open-angle), allele A, OR 1.48, p=7E-30, PMID 25173105]	Strong
SIX6	rs33912345	H141N	C/C	chip	C (2 copy/s)	unfavourable	H141N: the C allele (Asn141) thinnns the retinal nerve fibers layer; ~1.3× more glaucoma	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TXNRD2	rs35934224		C/C	chip	C (2 copy/s)	unfavourable	Mitochondrial thioredoxin reductase (antioxidant defense of ganglion cells) [GWAS Catalog: Glaucoma (primary open-angle), C allele, OR 1.28, p=4E-11, PMID 26752265]	Moderate
ATOH7	rs1900004		C/T	chip	T (1 copy/s)	without judgment	ATOH7 (development of ganglion cells): the T allele is associated with a smaller optical disc; it influences the reading of the excavation, without clear risk [GWAS Catalog: Optical disc area, T allele, beta -0.10]	Moderate

Macular degeneration of age

Complement (CFH, C3, CFB), ARMS2/HTRA1 and Bruch membrane lipids.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CFH	rs1061170	Y402H	C/T	chip	C (1 copy/s)	unfavourable (light)	Y402H: the C allele (His402) reduces the complement regulation in the macula; ~2.5× more DMI per copy	Strong
ARMS2/HTRA1	rs10490924	A69S	G/G	chip	T (0 copy/s)	favourable	A69S: the T allele is the second largest DMI factor (~2.7× per copy)	Strong
CFB	rs641153	R32Q	G/G	chip	A (0 copy/s)	neutral	R32Q: Allele A reduces complement activation and protects from DMI (~0.5×)	Strong
C3	rs2230199	R102G	G/C	chip	C (1 copy/s)	unfavourable (light)	C3 R102G (C3F allotype): more activation of the complement in the macula; ~1.4× more DMI per copy [GWAS Catalog: Macular degeneration, allele G (strand +) with protective sense]	Strong
CETP	rs3764261		C/C	chip	A (0 copy/s)	favourable	CETP promoter: A allele rises to HDL and is associated with ~1.15× plus DMI [GWAS Catalog: Age-related macular degeneration, allele A, OR 1.15]	Moderate
LIPC	rs10468017		C/C	chip	C (2 copy/s)	unfavourable (light)	Liver lipase (LIPC): T allele is associated with higher HDL and less DMI (~0.85×); C is the usual	Moderate
TIMP3	rs9621532		A/A	chip	A (2 copy/s)	unfavourable	Bruch membrane metalloproteinase inhibitor [GWAS Catalog: Age-related macular degeneration, allele A, OR 1.41, p=1E-11, PMID 20385819]	Moderate

Crystalline (cataracts)

Ionic transport and splicing in the lens.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SLC24A3	rs4814857		A/A	imputation DR2=0.97	G (0 copy/s)	favourable	Sodium-calcium-potassium exchanger: the strongest common locus of cataracts [GWAS Catalog: Cataracts, G allele, OR 1.09, p=2E-70, PMID 34127677]	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
RBFOX1	rs10500355	A/A	chip		A (2 copy/s)	unfavourable	Splicing regulator; cataracts [GWAS Catalog: Cataracts, allele A, OR 1.04, p=6E-26, PMID 34127677]	Moderate
SOX2-OT	rs9842371	C/C	imputation DR2=0.98		T (0 copy/s)	favourable	SOX2-OT locus of cataracts [GWAS Catalog: Cataracts, T allele, OR 1.09, p=3E-33, PMID 34127677]	Moderate

Myopia and refractive error

Retinal-sclera signaling that controls the axial growth of the eye.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
GJD2	rs634990	C/C	chip		T (0 copy/s)	favourable	36 connexin (retinal synapses): the largest common locus of myopia [GWAS Catalog: Refractive error, T allele, beta 0.057 [0.054-0.061] SD unit increase, p=2E-221, PMID 42009823]	Strong
LAMA2	rs12193446	A/G	chip		A (1 copy/s)	unfavourable (light)	Laminin α2 of sclera: elongation of the eye [GWAS Catalog: Refractive error, allele A, beta 0.113 [0.11-0.12] SD unit decrease, p=1E-300, PMID 42009823]	Strong
KCNQ5	rs7744813	A/A	chip		A (2 copy/s)	unfavourable	Retinal potassium channel: myopia [GWAS Catalog: Refractive error, allele A, beta 0.054 [0.05-0.057] SD unit decrease, p=6E-188, PMID 42009823]	Moderate
PRSS56	rs2741297	A/A	imputation DR2=0.99		A (2 copy/s)	unfavourable	Protease of the axial growth of the eye [GWAS Catalog: Refractive error, allele A, beta 0.046 [0.042-0.05] SD unit decrease, p=7E-118, PMID 42009823]	Moderate

5. GWAS loci burden

Sum of the loci of GWAS Catalog ($p \leq 510-8$) of each trait. Alleles are oriented towards more risk (or more pressure, excavation and myopia) and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Glaucoma	759 (720)	742 of 1440	717	91.8	unfavourable
Intraocular pressure	583 (515)	563 of 1030	513	93.4	unfavourable
Optical nerve (vertical excavation)	651 (569)	575 of 1138	568	38.1	typical
Macular degeneration	178 (164)	151 of 328	156	32.7	typical
Cataracts	107 (95)	74 of 190	95	74.9	typical
Myopia	106 (97)	106 of 194	97	99.5	unfavourable

6. Favourable

- **DMI:** ARMS2 A69S G/G and CETP C/C without risk alleles; 53.4, 71.7 and 41.1 scores; 32.7 loci. Moderate to strong (ARMS2).
- **MYOC Q368X absent** (G/G): without the main monogenic cause of juvenile or adult glaucoma. Strong for the variant.
- **LOXL1 G153D G/A:** 1 copy of the protective allele, less exfoliation syndrome than in G/G. **Moderate.**
- **TMCO1 A/A** and **AFAP1 A/A:** without alleles of pressure and glaucoma. **Moderate.**
- **Excavation of the optic nerve:** loci load at 38.1. **Weak.**
- **Cataracts:** SLC24A3 and SOX2-OT without risk alleles; typical loci load (74.9). **Weak.**

7. Unfavourable

- **Primary open angle glaucoma:** percentiles 97.9, 81.5 and 59.2; loci load at 91.8. **Moderate to strong.**
- **Intraocular pressure:** score in 94.0 (the other at 63.3) and loci load at 93.4; GAS7 A/A, CAV1/CAV2 and ABCA1 with 1 allele. **Moderate.**
- **Optical nerve:** CDKN2B-AS1 A/A, SIX6 C/C (Asn141) and TXNRD2 C/C (2 copies each); ATOH7 C/T. **Moderate.**
- **Myopia:** genomic diagnosis at 99.2, loci at 99.5, refractive error in 7.6 (myopic); KCNQ5 and PRSS56 A/A, LAMA2 1 allele. **Strong.**
- **Cataracts:** 86.6 (genomic) and 72.2; RBFOX1 A/A. **Moderate.**
- **CFH Y402H C/T** and **C3 R102G** (1 copy each), TIMP3 A/A (2 copies) and LIPC without the protective allele. Compensated by ARMS2 G/G. **Moderate.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Remodelling of the extracellular matrix	Trabecular mesh and sieve (LOXL1, CAV1, GAS7): drainage of the aqueous mood and resistance of the optic nerve; genetically high intraocular pressure	Unfavourable
Stiffness of the Extracellular Matrix	The stiffness of the sclera and the sieve with age increases the mechanical stress in the optic nerve; the myopia elongates and thinning the sclera	Unfavourable
Oxidative stress	TXNRD2 C/C (thioredoxin reductase of ganglion cells) and crystalline (cataracts at percentile 86.6)	Unfavourable (mild)
Cell Senescence	CDKN2B-AS1 A/A (senescence p15/p16 locus) in ganglion cells and trabecular mesh	Unfavourable (mild)
Chronic Inflammation	Complement of macula: CFH Y402H 1 copy, but ARMS2 G/G and DMI typical	Neutral
Protein Aggregation	Cataract is aggregation of crystals; exfoliation syndrome is aggregation of fibrils (LOXL1 with protective 1 allele)	Neutral
Vascular Aging	The perfusion of the optic nerve counts in normal pressure glaucoma; LDL and ApoB low and exercise already in the plan	Conditional

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Complete ophthalmological examination now and then every 1-2 years:** intraocular pressure (with pachymetry), OCT of the nerve fiber layer and the optic disc, and visual fields if there is doubt. It is the measure with the greatest impact of this report; the standards recommend shorter intervals in high-risk people, such as polygenic risk and myopia. *Suggestion (not applied).*
2. **To ask the doctor to register the genetic risk of glaucoma** in the process, to lower the threshold of treatment if the pressure or OCT begins to change. *Suggestion (not applied).*
3. **Corticosteroids:** in inhaled or prolonged oral eye drops, they rise the intraocular pressure in part of the people. If necessary, measure the pressure after 2–6 weeks. *Suggestion (not applied).*
4. **Myopia and retina:** observation of peripheral retina with dilatation. Go to urgency with flashes of light, rain of fly flies or a curtain in the visual field (retinal detachment, more frequent in myopia). *Suggestion (not applied).*
5. **Regular aerobic exercise** (low intraocular pressure) and avoid prolonged head-down positions and long apneas with effort. Caffeine in very high doses slightly rises the pressure in the predispositions. *Suggestion (not applied).*
6. **Macula and crystalline:** sun glasses with filter UV, green leaf vegetables (lutein and zeaxanthin), fish and omega-3 3.2–4 g/day (already in the plan), zero tobacco and controlled glycemia. AREDS2 supplements are only indicated if intermediate DMI appears. *Suggestion (not applied).*

10. Limitations

- The glaucoma has rare forms of high effect (MYOC in addition to Q368X, OPTN, TBK1) that the chip does not read completely.
- The 127 score variants at percentile 97.9 and the genomic one in the 59.2 disagree; the loci load (glaucoma and pressure above the 90th percentile) supports the high signal.
- Pressure and myopia scores predict trends; measurement in ophthalmologic examination always has priority.
- The load of loci GWAS sums SNPs in linkage imbalance; it is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). The absent SNPs come from local imputation (Beagle 5.5, public panel of complete genomes to 30x), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the imputed VCF; in the palindromic SNPs the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed. Alleles published with protective effect (negative beta or $OR < 1$) were reversed for the sense of risk.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog (12 scores: glaucoma, intraocular pressure, DMI, cataracts, refractive error and myopia). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), 6 sets (2 384 loci), alleles oriented by the sign of the published effect, and empirical calibration in the same reference population.
- **Interpretation:** Craig 2020 and Gharahkhani 2021 (glaucoma), MacGregor 2018 (intraocular pressure), Fritsche 2016 (DMI), Jiang 2021 and Jung 2025 (cataracts), Hysi 2020 and Clark 2023 (refractive error), AREDS2 2013, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Glaucoma: intraocular pressure and drainage	TMC01	rs4656461	A/A	chip	G	0	strand +	without the risk allele
Glaucoma: intraocular pressure and drainage	CAV1/CAV2	rs4236601	G/A	chip	A	1	strand +	G/A: 1 copy of allele A (higher intraocular pressure)
Glaucoma: intraocular pressure and drainage	ABCA1	rs2472493	G/A	chip	G	1	strand +	1 copy of risk allele
Glaucoma: intraocular pressure and drainage	GAS7	rs9913911	A/A	chip	A	2	strand +	2 risk allele copies
Glaucoma: intraocular pressure and drainage	AFAP1	rs4619890	A/A	chip	G	0	strand +	without the risk allele
Glaucoma: intraocular pressure and drainage	MYOC	rs74315329	G/G	chip	A	0	strand +	without the risk allele
Glaucoma: intraocular pressure and drainage	LOXL1	rs3825942	G/A	chip	G	1	strand +	G/A: 1 copy of allele A, the least common and protective; risk of exfoliation syndrome lower than in G/G (most)
Glaucoma: vulnerability of the optic nerve	CDKN2B-AS1 (9p21)	rs4977756	A/A	chip	A	2	strand +	2 risk allele copies
Glaucoma: vulnerability of the optic nerve	SIX6	rs33912345	C/C	chip	C	2	strand +	2 risk allele copies
Glaucoma: vulnerability of the optic nerve	TXNRD2	rs35934224	C/C	chip	C	2	strand +	2 risk allele copies
Glaucoma: vulnerability of the optic nerve	ATOH7	rs1900004	C/T	chip	T	1	strand +	C/T: 1 copy of the T allele (lightly smaller optical disc)
Macular degeneration of age	CFH	rs1061170	C/T	chip	C	1	strand +	1 copy of risk allele
Macular degeneration of age	ARMS2/HTRA1	rs10490924	G/G	chip	T	0	strand +	without the risk allele
Macular degeneration of age	CFB	rs641153	G/G	chip	A	0	strand +	without the protective allele (common genotype)
Macular degeneration of age	C3	rs2230199	G/C	chip	C	1	strand + (palindromic; assumed +)	G/C: 1 copy of the risk allele (C3 R102G, allotype C3F)
Macular degeneration of age	CETP	rs3764261	C/C	chip	A	0	strand +	C/C: without allele A (higher HDL) that is associated with more DMI

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Macular degeneration of age	LIPC	rs10468017	C/C	chip	C	2	without data	C/C: without the T allele (higher HDL) that is associated with less DMI
Macular degeneration of age	TIMP3	rs9621532	A/A	chip	A	2	strand +	2 risk allele copies
Crystalline (cataracts)	SLC24A3	rs4814857	A/A	imputation DR2=0.97	G	0	strand +	without the risk allele
Crystalline (cataracts)	RBFOX1	rs10500355	A/A	chip	A	2	strand + (palindromic; assumed +)	2 risk allele copies
Crystalline (cataracts)	SOX2-OT	rs9842371	C/C	imputation DR2=0.98	T	0	strand +	without the risk allele
Myopia and refractive error	GJD2	rs634990	C/C	chip	T	0	strand +	without the risk allele
Myopia and refractive error	LAMA2	rs12193446	A/G	chip	A	1	strand +	1 copy of risk allele
Myopia and refractive error	KCNQ5	rs7744813	A/A	chip	A	2	strand +	2 risk allele copies
Myopia and refractive error	PRSS56	rs2741297	A/A	imputation DR2=0.99	A	2	strand +	2 risk allele copies